# tinyAM :: CHEATSHEET

Formula-driven age-structured assessment

## 1 Setup tidy inputs

### Wrangle your tables, then build a list

```
library(tinyAM)
catch_data <- tidyr::pivot_longer(catch_wide,
  cols = starts_with("age_"), names_to = "age", names_prefix = "age_",
  names_transform = list(age = as.integer), values_to = "obs")
dat <- list(
  catch = catch_data, index = index_data,
  weight = weight_data, maturity = maturity_data,
  pop_covariates = population_data
)
check_tam_data(dat)
# Optional: dat <- as_tam_data(dat)
```

**catch**

| year | age | obs |
|---|---|---|
| 1983 | 2 | 18 |
| 1983 | 3 | 2585 |
| ⋮ | ⋮ | ⋮ |
| 2024 | 19 | 0 |
| 2024 | 20 | 0 |

**weight**

| year | age | obs |
|---|---|---|
| 1983 | 2 | 0.0704 |
| 1983 | 3 | 0.201 |
| ⋮ | ⋮ | ⋮ |
| 2024 | 19 | 12.9 |
| 2024 | 20 | 14.1 |

**maturity**

| year | age | obs |
|---|---|---|
| 1983 | 2 | 0.00003 |
| 1983 | 3 | 0.00486 |
| ⋮ | ⋮ | ⋮ |
| 2024 | 19 | 1.00000 |
| 2024 | 20 | 1.00000 |

**index**

| year | age | survey | samp_time | obs | q_block |
|---|---|---|---|---|---|
| 1983 | 2 | Fall 2J3KL | 0.8 | 698822 | [2,3) |
| 1983 | 3 | Fall 2J3KL | 0.8 | 517528 | [3,4) |
| ⋮ | ⋮ | ⋮ | ⋮ | ⋮ | ⋮ |
| 2024 | 19 | Fall 2J3KL | 0.8 | 0.0 | [6,21) |
| 2024 | 20 | Fall 2J3KL | 0.8 | 0.0 | [6,21) |

**pop_covariates (optional)**

| year | age | NCAM_M_assumption |
|---|---|---|
| 1983 | 2 | 0.43 |
| 1983 | 3 | 0.38 |
| ⋮ | ⋮ | ⋮ |
| 2024 | 19 | 0.36 |
| 2024 | 20 | 0.36 |

**Keys:** year + age; add survey for indices. Use complete catch/biology and population-covariate grids. Weight/maturity cannot be missing; catch/index observations may be NA.

**Example units:** thousands of fish; kg/fish for weight; proportions for maturity. Sampling/spawning times are year fractions. Older biological ages can be retained above the model plus group.

## 2 Specify competing model hypotheses

### An example baseline, not the package defaults

```
base <- fit_tam(
  dat, years = 1983:2024, ages = 2:14,
  N_settings = list(process = "off", init = "exp"),
  rec_settings = list(process = "iid", mu_form = ~ 1),
  F_settings = list(process = "ar1", mu_form = ~ 1),
  M_settings = list(process = "off", mu_supplied = ~ I(0.3)),
  catch_settings = list(sd_form = ~ 1),
  index_settings = list(sd_form = ~ 1, q_form = ~ q_block),
  silent = TRUE
)
```

### Change one assumption at a time

```
F_mono <- update(base,
  F_settings = list(mu_form = ~ mono(age)))
rec_bh <- update(base,
  rec_settings = list(mu_form = ~ bh(ssb)))
```

```
q_mono <- update(base,
  index_settings = list(q_form = ~ mono(q_block)))
N_iid <- update(base,
  N_settings = list(process = "iid"))
```

```
fits <- list(Base = base, F_curve = F_mono,
             BH = rec_bh, Mono_q = q_mono,
             N_process = N_iid)
```

Partial updates retain other settings. `mu_form`: means; `q_form`: catchability; `sd_form`: SDs. Use `check_tam()` before comparing fits; convergence does not establish identifiability.

## 3 Visualize and compare

### Explore fits and test forecasting skill

```
vis_tam(model_list = fits, output_file = "tinyAM_dashboard.html",
        open_file = FALSE)
```

```
hind <- lapply(fits, fit_hindcast, folds = 20)
```

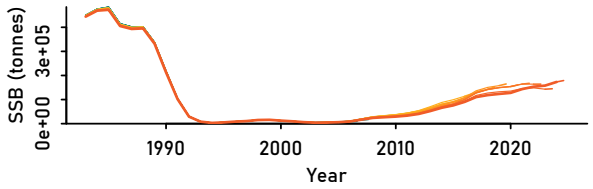
**Open the example dashboard**: Fishery → Numbers-at-age, then Residuals. Keep the companion HTML beside this PDF.

### One-year forecast log-RMSE

| Model | RMSE | Folds | Rank |
|---|---|---|---|
| Mono q | 0.676 | 20/20 | 1 |
| Base | 0.678 | 20/20 | 2 |
| F curve | 0.701 | 20/20 | 3 |
| BH | 0.707 | 19/20 | – |
| N process* | 0.798 | 2/20 | – |

Lower is better on the same observations. Only complete, converged runs are ranked. Folds count scored years. * Main fit did not converge.

### SSB stability across historical peels




Base-model SSB across historical peels. Forecast RMSE scores catch/survey observations, not SSB.

Experimental software; check assumptions and diagnostics before interpretation. Reproduce these examples with `cheatsheet/examples.R`. github.com/PaulRegular/tinyAM

# tinyAM :: MODEL REFERENCE

R specifications and the population assumptions they represent

## Abundance and cohort survival

```
N_settings = list(
  process = "iid", init = "exp")
```

$Z = F + M$; recruitment $R_t = N_{t,1}$. Age indices $1, \ldots, A$ run from the youngest modeled age to the plus group; N is start-of-year abundance.

$$\begin{aligned}\widehat{N}_{t,a} &= N_{t-1,a-1}e^{-Z_{t-1,a-1}},\\ \widehat{N}_{t,A} &= N_{t-1,A-1}e^{-Z_{t-1,A-1}}\\ &\quad + N_{t-1,A}e^{-Z_{t-1,A}},\\ \log N_{t,a} &= \log \widehat{N}_{t,a} + \eta^N_{t,a}.\end{aligned}$$

For $t \geq 2$, non-plus ages use the first survival equation. Initial older ages use `"exp"` survivorship, `"free"` fixed states, or `"random"` IID survivorship errors.

## Recruitment

```
rec_settings = list(process = "rw")
# Or a stock-recruit curve:
rec_settings = list(process="iid", mu_form=~bh(ssb))
```

$$\begin{aligned}\text{RW: } \log R_t - \log R_{t-1} &= \epsilon_t,\\ \log R_t &= \log g(SSB_{t-L}) + u_t,\\ g_{BH}(S) &= \alpha S/(1+\beta S),\\ g_{Ricker}(S) &= \alpha S e^{-\beta S}.\end{aligned}$$

$\alpha, \beta > 0$; $\epsilon_t \sim \mathcal{N}(0, \sigma_R^2)$. IID has $u_t = \epsilon_t$; AR1 has $u_t = \phi_R u_{t-1} + \epsilon_t$ and a stationary start. Curves describe medians. Lag $L$ defaults to the youngest biological age; early boundary R is free. No RW + curve.

## Biomass and spawning biomass

```
dat$maturity$spawn_time <- 0.25
```

$$SSB_t = \sum_b n_{t,b} w_{t,b} p_{t,b} e^{-\tau_{t,b} Z_{t,b}}.$$

Biomass is $\sum_b n_{t,b} w_{t,b}$. $b$: retained biological age; $n$: its abundance; $w$: weight; $p$: mature fraction; $\tau$: year fraction (default 0). Hidden ages retain biology/timing and share plus-age mortality.

## Mean fishing mortality

```
F_settings = list(process = "iid",
  mu_form = ~ logistic(age))
```

$$\begin{aligned}S(a) &= \{1 + e^{-k(a-a_{50})}\}^{-1}, \quad k > 0,\\ \log F_{t,a} &= \beta_F + \log S(a) + \eta^F_{t,a}.\end{aligned}$$

`mu_form = ~ mono(age)` instead adds cumulative nonnegative log-mean steps. Both curves require IID/AR1; annual F can depart from the rising mean.

## Natural mortality

```
M_settings = list(process = "iid",
  mu_supplied = ~ M0)
```

$$\log M_{t,a} = \log M^0_{t,a} + \eta^M_{t,a}.$$

`M0` comes from `pop_covariates`. M states share age blocks and start in year 2 by default; other cells retain the mean. `"off"` fixes M to its mean; `mu_form` adds log-mean effects.

## Variation in N, F and M

```
F_settings = list(process = "cor_rw",
  sd_form = ~ factor(age))
```

Ordinary age-based SD formulas give $\sigma_a = \exp(X_a \beta_\sigma)$. N, F and M have separate parameters; shared SDs do not share states.

$$\begin{aligned}\text{IID: } \eta_{t,a} &\sim \mathcal{N}(0, \sigma_a^2),\\ \text{RW: } \eta_{t,a} - \eta_{t-1,a} &\sim \mathcal{N}(0, \sigma_a^2).\end{aligned}$$

AR1 stationary covariance (state/block distances):

$$\text{Cov}(\eta_{t,a}, \eta_{s,b}) = \frac{\sigma_a \sigma_b \phi_t^{|t-s|} \phi_a^{|a-b|}}{(1-\phi_t^2)(1-\phi_a^2)}.$$

F `"cor_rw"`: independent annual increment vectors with

$$\Sigma_{ab} = \sigma_a \sigma_b \rho^{|a-b|}, \quad -1 < \rho < 1.$$

RW starts are unpenalized. N/M `"off"` gives $\eta = 0$.

## Survey catchability

```
index_settings = list(
  q_form = ~ mono(q_block), q_link = "log")
```

$$\eta_{q,k} = \beta_q + \sum_{j<k} d_j, \quad d_j \geq 0, \qquad q_k = e^{\eta_{q,k}}.$$

Zero steps give exact plateaus; `~ q_block` leaves age patterns unrestricted. With `q_link = "logit"`:

$$q_k = \{1 + e^{-\eta_{q,k}}\}^{-1}.$$

`~ logistic(age)` multiplies inverse-link q by $S(a)$. `iid()`, `rw()` and `ar1()` can add Gaussian variation. q covariates come from `index`.

## Catch and survey observations

```
catch_settings = list(sd_form = ~ 1)
index_settings = list(sd_form = ~ survey)
```

$$\begin{aligned}\widetilde{C}_{t,a} &= N_{t,a}\frac{F_{t,a}}{Z_{t,a}}(1 - e^{-Z_{t,a}}),\\ \tilde{I}_i &= q_i N_{t_i,a_i} e^{-s_i Z_{t_i,a_i}},\\ \log Y_i &\sim \mathcal{N}(\log \widetilde{Y}_i, \sigma_i^2),\\ \sigma_i &= \sigma_i^0 \exp(X_{sd,i}\beta_{sd}).\end{aligned}$$

$s_i$: survey sampling fraction. $\sigma_i^0 = 1$ unless a baseline SD is provided. Gaussian formula effects can also modify log SD. Predictions are medians; zero/NA observations are excluded, with complete predictions.

## Check fit, uncertainty and predictive skill

```
check_tam(fit)
hind <- fit_hindcast(fit, folds = 20)
hind$hindcast_rmse
```

The joint likelihood includes active process, formula-effect and observation densities. RTMB integrates random effects by Laplace approximation; intervals use local curvature. Standardized residuals $(\log Y_i - \log \widetilde{Y}_i)/\sigma_i$ depend on fitted, shrunken states. Convergence does not establish identifiability. Hindcasts return log RMSE and retrospective Mohn's $\rho$.

Equation details: `?"tinyAM-model"`, `?recruitment_formulas`, `?catchability_curves` · github.com/PaulRegular/tinyAM